

N+ - Network Fundamentals and building network

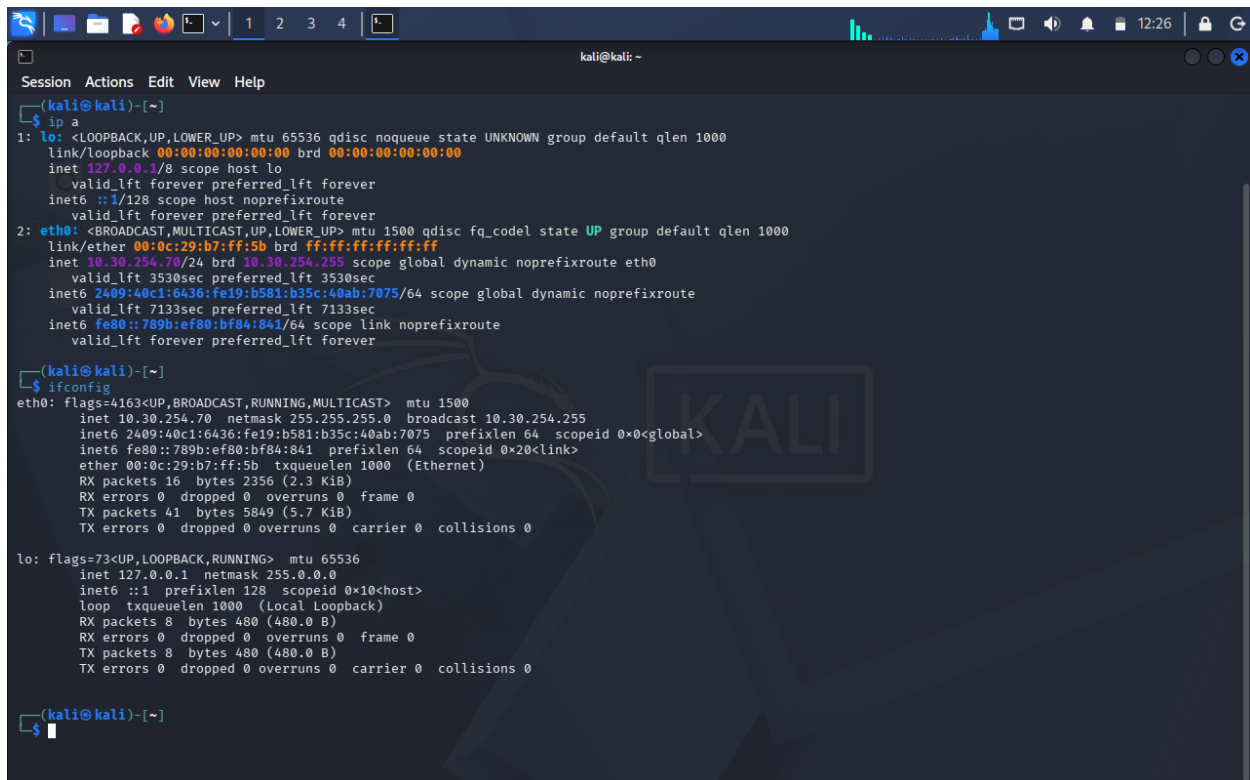
Configure IPv6 Addressing

[Q-1] Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Linux Commands

Ip a

If config



```
(kali@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b7:ff:5b brd ff:ff:ff:ff:ff:ff
    inet 10.30.254.70/24 brd 10.30.254.255 scope global dynamic noprefixroute eth0
        valid_lft 3530sec preferred_lft 3530sec
    inet6 2409:40c1:6436:fe19:b581:b35c:40ab:7075/64 scope global dynamic noprefixroute
        valid_lft 7133sec preferred_lft 7133sec
    inet6 fe80::789b:ef80:bf84:841/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

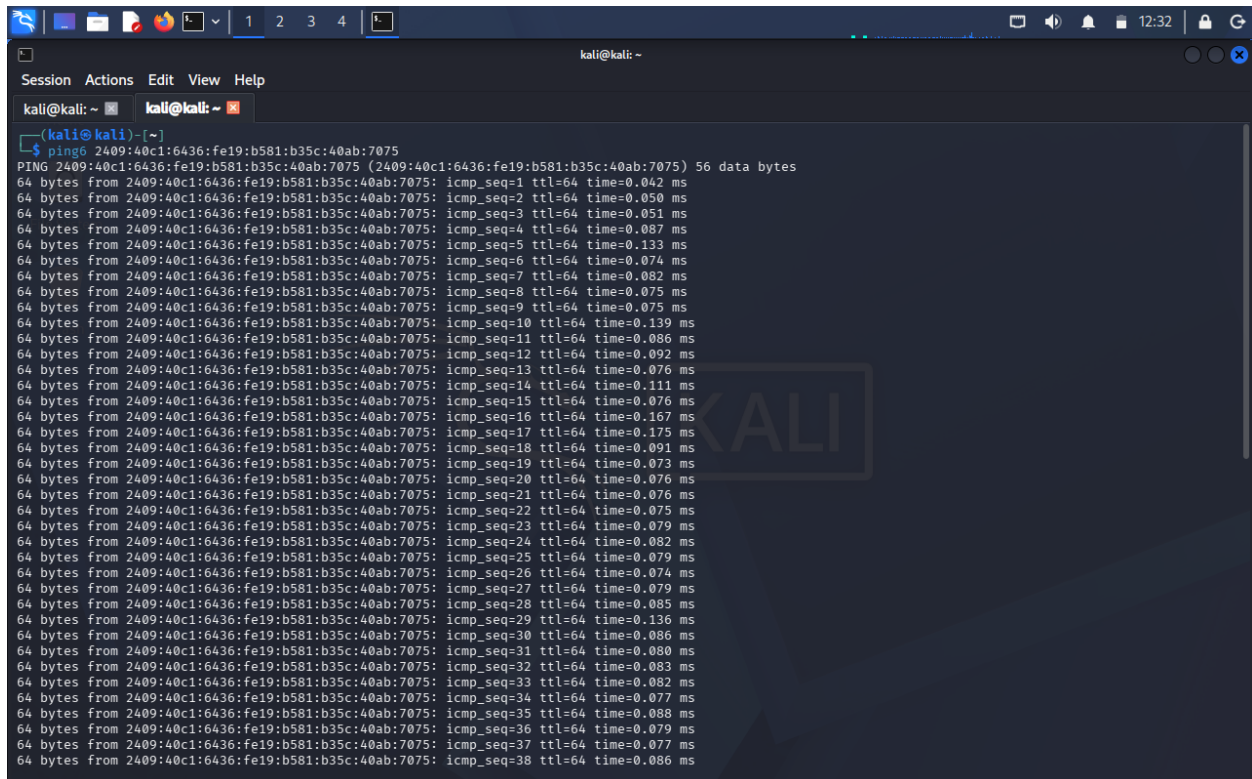
(kali@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.30.254.70 netmask 255.255.255.0 broadcast 10.30.254.255
    inet6 2409:40c1:6436:fe19:b581:b35c:40ab:7075 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::789b:ef80:bf84:841 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:b7:ff:5b txqueuelen 1000 (Ethernet)
    RX packets 16 bytes 2356 (2.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 41 bytes 5849 (5.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

[Q-2] Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.
Hint: Use the ping6 or ping -6 command for IPv6 testing.

Verification Commands

Ping6



```
kali@kali: ~  
$ ping6 2409:40c1:6436:fe19:b581:b35c:40ab:7075  
PING 2409:40c1:6436:fe19:b581:b35c:40ab:7075 (2409:40c1:6436:fe19:b581:b35c:40ab:7075) 56 data bytes  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=1 ttl=64 time=0.042 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=2 ttl=64 time=0.050 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=3 ttl=64 time=0.051 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=4 ttl=64 time=0.087 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=5 ttl=64 time=0.133 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=6 ttl=64 time=0.074 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=7 ttl=64 time=0.082 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=8 ttl=64 time=0.075 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=9 ttl=64 time=0.075 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=10 ttl=64 time=0.139 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=11 ttl=64 time=0.086 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=12 ttl=64 time=0.092 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=13 ttl=64 time=0.076 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=14 ttl=64 time=0.111 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=15 ttl=64 time=0.076 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=16 ttl=64 time=0.167 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=17 ttl=64 time=0.175 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=18 ttl=64 time=0.091 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=19 ttl=64 time=0.073 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=20 ttl=64 time=0.076 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=21 ttl=64 time=0.076 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=22 ttl=64 time=0.075 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=23 ttl=64 time=0.079 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=24 ttl=64 time=0.082 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=25 ttl=64 time=0.079 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=26 ttl=64 time=0.074 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=27 ttl=64 time=0.079 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=28 ttl=64 time=0.085 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=29 ttl=64 time=0.136 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=30 ttl=64 time=0.086 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=31 ttl=64 time=0.080 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=32 ttl=64 time=0.083 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=33 ttl=64 time=0.082 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=34 ttl=64 time=0.077 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=35 ttl=64 time=0.088 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=36 ttl=64 time=0.079 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=37 ttl=64 time=0.077 ms  
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=38 ttl=64 time=0.086 ms
```

Ping -6

```

kali@kali: ~
Session Actions Edit View Help
kali@kali: ~ kali@kali: ~ kali@kali: ~

(kali@kali)~$ ping -6 2409:40c1:6436:fe19:b581:b35c:40ab:7075
PING 2409:40c1:6436:fe19:b581:b35c:40ab:7075 (2409:40c1:6436:fe19:b581:b35c:40ab:7075) 56 data bytes
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=1 ttl=64 time=0.041 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=3 ttl=64 time=0.047 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=4 ttl=64 time=0.082 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=5 ttl=64 time=0.087 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=6 ttl=64 time=0.080 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=7 ttl=64 time=0.077 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=8 ttl=64 time=0.086 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=9 ttl=64 time=0.077 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=10 ttl=64 time=0.070 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=11 ttl=64 time=0.069 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=12 ttl=64 time=0.081 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=13 ttl=64 time=0.189 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=14 ttl=64 time=0.076 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=15 ttl=64 time=0.084 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=16 ttl=64 time=0.077 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=17 ttl=64 time=0.088 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=18 ttl=64 time=0.077 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=19 ttl=64 time=0.083 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=20 ttl=64 time=0.146 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=21 ttl=64 time=0.084 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=22 ttl=64 time=0.078 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=23 ttl=64 time=0.079 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=24 ttl=64 time=0.081 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=25 ttl=64 time=0.114 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=26 ttl=64 time=0.082 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=27 ttl=64 time=0.091 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=28 ttl=64 time=0.078 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=29 ttl=64 time=0.072 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=30 ttl=64 time=0.081 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=31 ttl=64 time=0.078 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=32 ttl=64 time=0.083 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=33 ttl=64 time=0.078 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=34 ttl=64 time=0.077 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=35 ttl=64 time=0.077 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=36 ttl=64 time=0.121 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=37 ttl=64 time=0.084 ms
64 bytes from 2409:40c1:6436:fe19:b581:b35c:40ab:7075: icmp_seq=38 ttl=64 time=0.070 ms
  
```

[Q-3] Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.
 Constraint: Document the steps and the observed address behavior in a text file.

sudo sysctl -w net.ipv6.conf.all.use_tempaddr=2

check address

ip -6 addr show

```

kali@kali: ~
Session Actions Edit View Help
kali@kali: ~ kali@kali: ~ kali@kali: ~ kali@kali: ~
(kali@kali)-[~]
$ sudo sysctl -w net.ipv6.conf.all.use_tempaddr=2
[sudo] password for kali:
net.ipv6.conf.all.use_tempaddr = 2
(kali@kali)-[~]
$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 2001:40c1:6436:f19:b581:b35c:40ab:7075/64 scope global dynamic noprefixroute
        valid_lft 6900sec preferred_lft 6900sec
    inet6 fe80::789b:ef84:bf84:841/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(kali@kali)-[~]
$
  
```

[Q-4] Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Restaurant names and IPv6 addresses

```

restaurants = {
    "Pizza Palace": "2001:db8::1",
    "Burger House": "2001:db8::2",
    "Spice Garden": "2001:db8::3",
    "Food Corner": "2001:db8::4",
    "Tasty Bites": "2001:db8::5"
}
  
```

```
search = input("Enter part of the restaurant name: ").lower()
```

```
for name, ipv6 in restaurants.items():  
    if search in name.lower():  
        print(f"Restaurant: {name}")  
        print(f"IPv6 Address: {ipv6}")  
        found = True  
  
    print("No matching restaurant found.")
```

Example Run

Input:

pizza

Output:

Restaurant: Pizza Palace
IPv6 Address: 2001:db8::1

[Q-5] Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Configuration Command

```
Sudo ip -6 addr add 2001:db8:acad:3::100/64 dev eth0
```

Verification

```
Ip -6 addr show eth0
```